

Relativistic hydrogenic transition rates in hydrocal

AI generated and edited by S. Schippers

September 22, 2026

1 Dirac hydrogenic wavefunctions

The Dirac spinor for a hydrogenic state with principal quantum number n and angular quantum number κ (where $\kappa = l$ for $j = l - \frac{1}{2}$ and $\kappa = -(l + 1)$ for $j = l + \frac{1}{2}$) is

$$\psi_{n\kappa m}(\mathbf{r}) = \frac{1}{r} \begin{pmatrix} P_{n\kappa}(r) \chi_{\kappa m}(\hat{r}) \\ i Q_{n\kappa}(r) \chi_{-\kappa m}(\hat{r}) \end{pmatrix}, \quad (1)$$

with the normalisation $\int_0^\infty (P^2 + Q^2) dr = 1$. The radial functions are

$$\begin{aligned} P_{n\kappa}(r) &= C \sqrt{1 + \varepsilon} (2\lambda r)^\gamma e^{-\lambda r} F(r), \\ Q_{n\kappa}(r) &= C \sqrt{1 - \varepsilon} (2\lambda r)^\gamma e^{-\lambda r} G(r), \end{aligned} \quad (2)$$

where $F(r)$ and $G(r)$ are polynomials of degree $n_r = n - |\kappa|$. The parameters are

$$\gamma = \sqrt{\kappa^2 - (\alpha Z)^2}, \quad (3)$$

$$N' = n_r + \gamma, \quad (4)$$

$$\lambda = Z/N', \quad (5)$$

$$\varepsilon = [1 + (\alpha Z/N')^2]^{-1/2}, \quad (6)$$

$$C = \text{normalisation constant}, \quad (7)$$

with α the fine-structure constant and Z the nuclear charge.

The normalisation constant C follows from $\int_0^\infty [P^2 + Q^2] dr = 1$. Writing $F(r) = \sum_i a_i r^i$ and $G(r) = \sum_i b_i r^i$ with polynomial degree n_r , the integral is evaluated termwise:

$$\int_0^\infty [P^2 + Q^2] dr = C^2 \sum_{i=0}^{n_r} \sum_{j=0}^{n_r} [(1 + \varepsilon) a_i a_j + (1 - \varepsilon) b_i b_j] \frac{\Gamma(2\gamma + i + j + 1)}{(2\lambda)^{i+j+1}}, \quad (8)$$

where the denominator follows from $\int_0^\infty (2\lambda r)^{2\gamma} r^{i+j} e^{-2\lambda r} dr = \Gamma(2\gamma + i + j + 1) / (2\lambda)^{i+j+1}$. Hence $C = [S]^{-1/2}$ with S the double sum in (8).

All radial integrals below are evaluated analytically by expanding the polynomials and integrating termwise using

$$\int_0^\infty r^p e^{-(\lambda_1 + \lambda_2)r} dr = \frac{\Gamma(p + 1)}{(\lambda_1 + \lambda_2)^{p+1}}. \quad (9)$$

2 Transition rates

All three multipole rates are expressed in a common form. For an electric 2^k -pole transition the rate in the long-wavelength approximation is

$$A_{\text{Ek}} = \frac{2(k + 1)}{k[(2k + 1)!!]^2} \alpha^3 \omega^{2k+1} (2j_f + 1) [3j(j_f, k, j_i; -\frac{1}{2}, 0, \frac{1}{2})]^2 |\langle l_f \| C^{(k)} \| l_i \rangle|^2 |R_k^{(\text{E})}|^2, \quad (10)$$

where the radial integral is

$$R_k^{(E)} = \int_0^\infty r^k [P_f P_i + Q_f Q_i] dr. \quad (11)$$

For a magnetic 2^k -pole,

$$A_{Mk} = \frac{2(k+1)}{k[(2k+1)!!]^2} \alpha^3 \omega^{2k+1} (2j_f + 1) [3j(j_f, k, j_i; -\frac{1}{2}, 0, \frac{1}{2})]^2 |R_k^{(M)}|^2, \quad (12)$$

but the radial integral involves the small components,

$$R_k^{(M)} = \int_0^\infty r^k [P_f Q_i + Q_f P_i] dr, \quad (13)$$

and the angular factor uses κ -dependent parity selectors $\Pi(\kappa_f, -\kappa_i; k)$ described below.

In both cases ω is the transition energy in Hartree, $\omega = |E_i - E_f|/(2\text{Ryd})$.

2.1 Electric dipole (E1)

For $k = 1$ (E1) the prefactor is

$$\frac{2(1+1)}{1 \cdot (3!!)^2} = \frac{4}{9},$$

the orbital reduced matrix element is

$$|\langle l_f \| C^{(1)} \| l_i \rangle|^2 = (2l_i + 1)(2l_f + 1) [3j(l_f, 1, l_i; 0, 0, 0)]^2,$$

and the radial integral is $R_1^{(E)}$:

$$A_{E1} = \frac{4}{3} \alpha^3 \omega^3 (2j_f + 1) [3j(j_f, 1, j_i; -\frac{1}{2}, 0, \frac{1}{2})]^2 \left| \int_0^\infty r [P_f P_i + Q_f Q_i] dr \right|^2. \quad (14)$$

(Here $\frac{4}{9} \times (2l_i + 1)(2l_f + 1)[3j(l_f, 1, l_i; 0, 0, 0)]^2$ reduces to the factor $4/3$ after evaluating the Gaunt coefficient.)

Selection rules:

- $\Delta l = \pm 1$ (parity change);
- $\Delta j = 0, \pm 1$ with $0 \rightarrow 0$ forbidden;
- $\kappa_f = -\kappa_i$ (e.g. $s_{1/2} \leftrightarrow p_{1/2}$) or $\kappa_f = \kappa_i \pm 1$ (e.g. $p_{1/2} \leftrightarrow d_{3/2}$).

2.2 Fully retarded E1 rate

Equation (14) is the long-wavelength (dipole) limit of the exact E1 rate. It is accurate only while the photon momentum $k = \omega/c$ is small compared with the momentum of the radiating electron. At $Z = 92$, $k \sim 0.2$ a.u. for the $2p \rightarrow 1s$ transition, and Eq. (14) overestimates the $2p_{3/2} \rightarrow 1s$ rate by $\sim 24\%$ while underestimating the $2p_{1/2} \rightarrow 1s$ rate by $\sim 8\%$. `dirac_transrate()` therefore evaluates the fully retarded four-component plane-wave matrix element

$$M = \langle a | \boldsymbol{\alpha} \cdot \hat{\boldsymbol{\varepsilon}} e^{-i\mathbf{k} \cdot \mathbf{r}} | b \rangle, \quad (15)$$

expanded in spherical partial waves $e^{-i\mathbf{k} \cdot \mathbf{r}} = 4\pi \sum_{L,m} (-i)^L j_L(kr) Y_{Lm}^*(\hat{k}) Y_{Lm}(\hat{r})$, and summed over the magnetic substates of both states and the two photon polarisations,

$$A = \frac{2k}{2j_b + 1} \sum_{m_a, m_b, \hat{\boldsymbol{\varepsilon}}} |M|^2 \quad (\text{atomic units}), \quad (16)$$

with $k = \omega/c$ the photon wavenumber and ω the transition energy in Hartree. The angular factor for each pair of spinor harmonics is reduced by `ang_int()` to coefficients of $j_L(kr)$ via $\int d\Omega Y_{l_1 m_1}^* Y_{l_2 m_2} e^{-ikr \cos \theta} = \sum_L c_L(l_1, m_1, l_2, m_2) j_L(kr)$; the largest order required is $L_{\max} = \max[l(\kappa_a) + l(-\kappa_b), l(-\kappa_a) + l(\kappa_b)]$. The radial factors are the cross-type integrals

$$R_+[L] = \int_0^\infty P_a(r) Q_b(r) j_L(kr) dr, \quad (17)$$

$$R_-[L] = \int_0^\infty Q_a(r) P_b(r) j_L(kr) dr, \quad (18)$$

evaluated by trapezoidal quadrature on the logarithmic grid $r \in [10^{-10}, 80]$ a.u. with 160000 points. The spherical Bessel functions j_L are computed in closed form for $L \leq 4$, by a power series for $L > 4$ and $kr < 8$, and by upward recursion otherwise. The j_L expansion of the exponential is essential: it introduces the retardation that the dipole approximation of Eq. (14) drops.

2.3 Electric quadrupole (E2)

`dirac_e2_rate()` returns the fully retarded plane-wave rate of Section 2.2 for parity-conserving pairs with $\Delta l = \pm 2$. For such pairs E2 is the leading multipole; higher multipoles are suppressed by $(\alpha Z)^2$. In the long-wavelength limit the rate reduces to the familiar formula

$$A_{E2} = \frac{\alpha^3 \omega^5}{75} (2j_f + 1) [3j(j_f, 2, j_i; -\frac{1}{2}, 0, \frac{1}{2})]^2 |\langle l_f \| C^{(2)} \| l_i \rangle|^2 \left| \int_0^\infty r^2 [P_f P_i + Q_f Q_i] dr \right|^2, \quad (19)$$

where the orbital reduced matrix element is $\langle l_f \| C^{(2)} \| l_i \rangle = (2l_i + 1)(2l_f + 1) [3j(l_f, 2, l_i; 0, 0, 0)]^2$. The analytic length-form evaluation of this expression was found to be buggy (a normalisation error made the $3d_{3/2} \rightarrow 1s$ rate at $Z = 1 \sim 3756\times$ too large), so the retarded expression of Section 2.2 is used instead, which reproduces the $3d \rightarrow 1s$ rates of Jitrik & Bunge [4] (see Section 2.7).

Selection rules:

- $\Delta l = \pm 2$ (parity unchanged);
- $\Delta j = 0, \pm 1, \pm 2$ with $0 \rightarrow 0, 0 \rightarrow 1, 1 \rightarrow 0$ forbidden;
- pairs with $\Delta l = 0$ return zero from `dirac_e2_rate()` and are covered by the M1 rate instead.

2.4 Magnetic dipole (M1)

`dirac_m1_rate()` covers two physically distinct M1 mechanisms.

Retarded plane-wave rate (different n). For $\Delta l = 0$ pairs with $n_i \neq n_f$ (e.g. $2s_{1/2} \rightarrow 1s_{1/2}$) the M1 rate is the fully retarded plane-wave rate of Section 2.2. The transition is driven by the small components of the Dirac wavefunctions; the long-wavelength cross-radial form

$$A_{M1} = \frac{\alpha^3 \omega^3}{3} (2j_f + 1) [3j(j_f, 1, j_i; -\frac{1}{2}, 0, \frac{1}{2})]^2 \left| \int_0^\infty [P_f Q_i + Q_f P_i] dr \right|^2 \quad (20)$$

is the leading-order limit.

Fine-structure M1 (same n). For the two fine-structure partners of the same (n, l) doublet, $j = l + \frac{1}{2} \rightarrow j' = l - \frac{1}{2}$ ($\Delta l = 0$, $\Delta j = 1$), the magnetic-dipole operator is $\boldsymbol{\mu} = -\mu_B(\mathbf{L} + 2\mathbf{S})$. $\mathbf{J}$ couples only equal- j states, so the transition is driven by the electron spin alone. The reduced matrix element is

$$\frac{|\langle j' \| L + 2S \| j \rangle|^2}{2j + 1} = \frac{l}{2l + 1},$$

and the rate is computed with the standard magnetic-dipole formula (in SI units),

$$A_{\text{M1}}^{\text{fs}} = \frac{4}{3} \frac{\omega^3 \mu_B^2}{\hbar c^3} \frac{l}{2l + 1}. \quad (21)$$

The retarded plane-wave integral of Section 2.2 cannot resolve this rate: at the very small photon momentum of the near-degenerate fine-structure splitting the partial-wave expansion suffers catastrophic cancellation (it returns numerically meaningless values even at $Z = 92$). Equation (21) reproduces the hydrogen $2p_{3/2} \rightarrow 2p_{1/2}$ rate $A \approx 4.4 \times 10^{-6} \text{ s}^{-1}$ ($\tau \approx 2.6$ days) for the Dirac fine-structure splitting of 10.97 GHz.

Selection rules (both mechanisms):

- $\Delta l = 0$ (same parity);
- $\Delta j = 0, \pm 1$ with $0 \rightarrow 0$ forbidden.

2.5 Combined rate

The function

$$A_{\text{total}}(n_i, \kappa_i \rightarrow n_f, \kappa_f) = A_{\text{E1}} + A_{\text{E2}} + A_{\text{M1}}$$

returns the total spontaneous emission rate in s^{-1} . Although E1 typically dominates whenever it is allowed, all three multipoles are summed explicitly. This is safe because the three selection rules are mutually exclusive: E1 requires a parity change ($l_i + l_f$ odd), E2 requires $l_i + l_f$ even with $\Delta l = \pm 2$, and M1 requires $\Delta l = 0$. No pair is therefore double-counted, and for parity-conserving pairs the sum reproduces the fully retarded plane-wave rate of Section 2.2 (up to the analytic fine-structure replacement for same- n M1). Summing remains correct for very highly charged ions, where the relativistic corrections make the smaller multipoles relatively more important.

2.6 Lifetime

The lifetime of a state (n, κ) is

$$\frac{1}{\tau_{n\kappa}} = \sum_{n_p=1}^n \sum_{\kappa_p} A_{\text{total}}(n, \kappa \rightarrow n_p, \kappa_p),$$

summing over all lower-energy states including same- n fine-structure partners. Unphysical transitions are excluded by checking $E_{\text{upper}} < E_{\text{lower}}$ and $l < n$ (the `valid_kappa()` condition).

2.7 Validation against literature

The point-nucleus relativistic E1 rates of `dirac_transrate()` were compared with independent fully relativistic calculations over a wide range of Z .

Pal'chikov (1998). Table 1 lists the twelve $n = 3 \rightarrow n = 2$ rates for $Z = 30$ and $Z = 92$ together with the values of Pal'chikov [1], Table V. The ratio $A_{\text{hydrocal}}/A_{\text{Pal}}$ lies in the interval $[0.999993, 1.000028]$: the two independent calculations agree to four significant figures or better, with no systematic offset.

Transition	Z	$A_{\text{Pal}} \text{ (s}^{-1}\text{)}$	$A_{\text{hydrocal}} \text{ (s}^{-1}\text{)}$
$3s_{1/2} \rightarrow 2p_{1/2}$	92	2.1137×10^{14}	2.11366×10^{14}
$3s_{1/2} \rightarrow 2p_{3/2}$	92	8.3304×10^{14}	8.33041×10^{14}
$3p_{1/2} \rightarrow 2s_{1/2}$	92	2.3288×10^{15}	2.32882×10^{15}
$3p_{3/2} \rightarrow 2s_{1/2}$	92	1.1988×10^{15}	1.19883×10^{15}
$3d_{3/2} \rightarrow 2p_{1/2}$	92	4.6390×10^{15}	4.63897×10^{15}
$3d_{3/2} \rightarrow 2p_{3/2}$	92	7.4781×10^{14}	7.47806×10^{14}
$3s_{1/2} \rightarrow 2p_{1/2}$	30	1.7587×10^{12}	1.75868×10^{12}
$3s_{1/2} \rightarrow 2p_{3/2}$	30	3.8733×10^{12}	3.87334×10^{12}
$3p_{1/2} \rightarrow 2s_{1/2}$	30	1.8799×10^{13}	1.87987×10^{13}
$3p_{3/2} \rightarrow 2s_{1/2}$	30	1.7958×10^{13}	1.79584×10^{13}
$3d_{3/2} \rightarrow 2p_{1/2}$	30	4.4547×10^{13}	4.45474×10^{13}
$3d_{3/2} \rightarrow 2p_{3/2}$	30	8.7049×10^{12}	8.70496×10^{12}

Table 1: E1 rates (s^{-1}) of `dirac_transrate()` compared with Pal’chikov [1], Table V.

Finite nuclear size. Knyazeva *et al.* [2] tabulate the $2p_{1/2} \rightarrow 1s$ and $2p_{3/2} \rightarrow 1s$ rates for one-electron ions for a point-like nucleus (W_0) and for a Fermi nuclear charge distribution ($W^{(e)}$). The point-nucleus values agree exactly with ours, Table 2; the finite-size correction is negligible at low Z and reaches $\sim 0.1\%$ at $Z = 92$. It is not included in the hydrogenic wavefunctions used here.

Z	Transition	W_0	ours	$W^{(e)}$
1	$2p_{1/2} \rightarrow 1s$	6.26835×10^8	6.26835×10^8	6.26835×10^8
1	$2p_{3/2} \rightarrow 1s$	6.26824×10^8	6.26824×10^8	6.26824×10^8
92	$2p_{1/2} \rightarrow 1s$	4.72601×10^{16}	4.72601×10^{16}	4.72033×10^{16}
92	$2p_{3/2} \rightarrow 1s$	3.95022×10^{16}	3.95022×10^{16}	3.94939×10^{16}

Table 2: $2p \rightarrow 1s$ rates (s^{-1}): point-like (W_0) and finite-nucleus ($W^{(e)}$) values of Knyazeva *et al.* [2] compared with `dirac_transrate()`.

High partial waves and zeros. The retarded operator is not truncated in L . The $5g_{7/2} \rightarrow 4f_{7/2}$ transition exercises j_L up to $L = 8$, and `dirac_transrate()` returns $1.52017 \times 10^5 \text{ s}^{-1}$, a physically expected hydrogenic rate. Selection-rule violations return exactly zero, e.g. $\Delta j = 2$ ($5g_{9/2} \rightarrow 4f_{5/2}$, $3d_{5/2} \rightarrow 2p_{1/2}$), the parity-forbidden $\Delta l = 2$ pair $4f_{5/2} \rightarrow 3p_{3/2}$, degenerate same- n , same- j pairs with $\omega = 0$ ($3d_{3/2} \rightarrow 3p_{3/2}$), and states with $l \geq n$ (e.g. $1p$).

M1 $2s_{1/2} \rightarrow 1s_{1/2}$. Sapirstein, Pachucki & Cheng [3] tabulate the lowest-order one-photon $2s_{1/2} \rightarrow 1s$ rate at $Z = 30$ as $3.7551 \times 10^{-8} \text{ a.u.}$ ($= 1.5524 \times 10^9 \text{ s}^{-1}$); `dirac_m1_rate()` returns $1.552527 \times 10^9 \text{ s}^{-1}$ (agreement $\sim 0.01\%$). Knyazeva *et al.* [2] list the sum of the M1 and two-photon rates at $Z = 92$ as $1.96240 \times 10^{14} \text{ s}^{-1}$; the M1-only value of `dirac_m1_rate()`, $1.946813 \times 10^{14} \text{ s}^{-1}$, is consistent (the two-photon contribution is small). At $Z = 1$ the code gives $2.495917 \times 10^{-6} \text{ s}^{-1}$, the well-known hydrogen value.

E2 $3d \rightarrow 1s$. For $Z = 1$ the hydrogenic E2 rates of Jitrik & Bunge [4] are 593.75 s^{-1} ($3d_{3/2} \rightarrow 1s$) and 593.74 s^{-1} ($3d_{5/2} \rightarrow 1s$). `dirac_e2_rate()` returns 5.940759×10^2 and $5.940601 \times 10^2 \text{ s}^{-1}$ (agreement $\sim 0.05\%$). At $Z = 92$ the $3d \rightarrow 2s$ E2 rate is $5.545458 \times 10^{13} \text{ s}^{-1}$ ($3d_{3/2} \rightarrow 2s_{1/2}$) and $5.780657 \times 10^{13} \text{ s}^{-1}$ ($3d_{5/2} \rightarrow 2s_{1/2}$). (The values 5.509576×10^{13} and $5.799182 \times 10^{13} \text{ s}^{-1}$ quoted in earlier versions of this document were produced by erroneous closed-form expressions

for j_3 and j_4 in the spherical Bessel routine; the corrected spherical Bessel functions give the values listed here.)

Fine-structure M1. For the $2p_{3/2} \rightarrow 2p_{1/2}$ fine-structure transition at $Z = 1$, `dirac_m1_rate()` returns $4.380826 \times 10^{-6} \text{ s}^{-1}$ ($\tau = 2.6$ days) from the analytic magnetic-dipole formula, Eq. (21), using the Dirac fine-structure splitting. The retarded plane-wave integral cannot be trusted for these near-degenerate pairs, which is why the analytic branch is used for all Z .

3 Relativistic radiative recombination cross section

The RR cross section for capture into *one specific magnetic substate* of a hydrogenic state (n, κ) at electron kinetic energy E_{kin} is obtained via the Milne relation,

$$\sigma_{\text{RR}}^{\text{vac}}(n, \kappa; E_{\text{kin}}) = \frac{\pi^2 \alpha a_0^2}{E_{\text{kin}}} \frac{g_{\text{bound}}}{2} \left. \frac{df}{dE} \right|_{E_{\text{kin}}}, \quad g_{\text{bound}} = 1, \quad (22)$$

where α is the fine-structure constant, a_0 the Bohr radius, and df/dE the bound-free oscillator strength per unit energy (Eq. 23). `dirac_rr_xsec()` implements $\sigma_{\text{RR}}^{\text{vac}}$ directly, with $g_{\text{bound}} = 1$: the explicit $(2j+1)$ degeneracy of the bound level has already been divided back out when forming df/dE from the reduced dipole matrix element (see below), so including it again here would double count it. The fully-retarded path of Section 3.5 keeps the same per-vacancy convention (its cross-section formula divides by $2j_b + 1$). This "per-vacancy" convention is what agrees with the tabulated Ichihara & Eichler cross sections (Section 3.7) and with `sigma_IchiharaEichlerRR()` (Section 4), both of which are also per-substate. A caller that needs the cross section for capture into the fine-structure *level as a whole* (i.e. summed over all $2j + 1$ magnetic substates – for example to compare against the non-relativistic full-subshell result below, or to build the level-population source term of Section 5.3) must multiply $\sigma_{\text{RR}}^{\text{vac}}(n, \kappa; E_{\text{kin}})$ by $(2j + 1)$ explicitly.

3.1 Bound-free oscillator strength

The dipole path described here (Eq. 24 and Section 3.4) is the *long-wavelength (electric-dipole)* route: it is the $\omega r/c \ll 1$ limit of the retarded multipole operators and omits the higher multipoles (E2, M1, ...) entirely, so it is reliable only while $\omega r/c \ll 1$ over the volume in which the bound-free overlap is significant (photon energies well below roughly $c/z \times$ (inverse bound-state radius), i.e. for hydrogenic ions from ~ 10 keV at $Z = 110$ up to ~ 50 keV at $Z = 1$). At higher energies the fully retarded, all-multipole calculation of Ichihara & Eichler (Section 3.7) grows relative to the dipole value – e.g. a factor ~ 500 at $Z = 110$, $n = 3$, 2 MeV – because retardation and the higher multipoles set in. `dirac_rr_xsec()` closes this gap with the fully-retarded plane-wave bound-free matrix element of Section 3.5, which is the default path; the dipole oscillator-strength machinery below survives as the legacy and validation route (and defines the non-relativistic limit).

For electric-dipole photoionisation the oscillator-strength density is

$$\frac{df}{dE} = \frac{2}{3} \frac{(2j_b + 1)(2j_c + 1)}{2j_b + 1} \left[3j(j_b, 1, j_c; -\frac{1}{2}, 0, \frac{1}{2}) \right]^2 |\langle l_b \| C^{(1)} \| l_c \rangle|^2 |R_{\text{bf}}|^2, \quad (23)$$

with the radial matrix element

$$R_{\text{bf}} = \int_0^\infty r [P_b(r)P_c(r) + Q_b(r)Q_c(r)] dr. \quad (24)$$

Here (P_b, Q_b) are the bound-state Dirac radial components normalised to unity, and (P_c, Q_c) are the continuum Dirac radial components normalised per unit energy (see below). The angular factors are the same as for the bound-bound E1 rate (Eq. 14) with the bound state as the upper (final) state and the continuum as the lower (initial) state.

The oscillator-strength density is summed over the E1-coupled continuum channels $\kappa_c \in \{\kappa + 1, \kappa - 1, -\kappa\}$; the third channel is required because E1 also couples states with the same $|\kappa|$ but opposite sign (e.g. $s_{1/2} \leftrightarrow p_{1/2}$), so capture into a bound $p_{1/2}$ state receives contributions from both $d_{3/2}$ and $s_{1/2}$ continuum partial waves. Channels of wrong parity or with $|\Delta j| > 1$ are dropped by the angular checks, and non-existent bound states ($l \geq n$) are rejected via the `valid_kappa()` condition.

3.2 Continuum Dirac wavefunction

The radial Dirac equation for a continuum state with energy $E = E_{\text{kin}} + m_e c^2$ is

$$\frac{dP}{dr} = -\frac{\kappa}{r} P + \frac{E + m_e c^2 - V(r)}{\hbar c} Q, \quad (25)$$

$$\frac{dQ}{dr} = +\frac{\kappa}{r} Q - \frac{E - m_e c^2 - V(r)}{\hbar c} P, \quad (26)$$

with $V(r) = -\alpha Z/r$ (point nucleus), in atomic units with $c = 1/\alpha$ and E_{kin} in Hartree. The system is integrated outward from a start radius $r_0 = 10^{-6}$ using an adaptive, error-controlled integrator (a DOPRI5 stepper, as implemented in `boost::numeric::odeint`, with absolute and relative tolerances 10^{-30} and 10^{-10} ; without Boost, a fixed-step RK4 sized to resolve both the near-origin $1/r$ curvature and the asymptotic oscillation period $2\pi/k$ is used instead). The starting values are obtained from a two-term Frobenius expansion near the origin, $P(r) \propto r^\gamma(1 + p_1 r)$ and $Q(r) \propto r^\gamma(q_0 + q_1 r)$, with

$$\gamma = \sqrt{\kappa^2 - (\alpha Z)^2}, \quad q_0 = \frac{c(\gamma + \kappa)}{Z}, \quad (27)$$

and the next-order coefficients p_1, q_1 fixed by matching the $\mathcal{O}(r^\gamma)$ terms of the ODE. (The leading-order-only condition is numerically degenerate for $\kappa < 0$ as $Z\alpha \rightarrow 0$, where $\gamma \rightarrow |\kappa| = -\kappa$; the next-order term removes that degeneracy.)

Because the raw functions P, Q span many orders of magnitude between the origin ($\sim r^\gamma$, down to $\sim 10^{-36}$ for $\gamma \sim 6$ at $r_0 = 10^{-6}$) and the far field ($\sim$ the asymptotic amplitude), no single absolute-error floor for the stepper can resolve the near-origin transient and yet stay meaningful at large r . `integrate_PQ()` therefore uses a hybrid representation: while the state is tiny it integrates the rescaled variables $p = P/r^\gamma, q = Q/r^\gamma$ — which are $\mathcal{O}(1)$ at r_0 for every κ , so the absolute tolerance acts on a well-conditioned state — and switches to the raw variables once $|P|, |Q| > 10^{-25}$, where the relative tolerance governs and the raw state stays $\mathcal{O}(\text{amplitude})$ in the far field instead of collapsing like $1/r^\gamma$. (Section 3.7 describes the runaway high- ℓ cross sections this change prevents.)

The integration proceeds to a large radius r_{max} where the potential is negligible. The asymptotic normalisation to the energy continuum,

$$\int_0^\infty [P_E P_{E'} + Q_E Q_{E'}] dr = \delta(E - E'), \quad (28)$$

is imposed by matching the large- r numerical solution to the asymptotic form

$$P_{\text{as}}(r) = \sqrt{\frac{2}{\pi k}} \sin(kr + \delta_\kappa), \quad (29)$$

$$Q_{\text{as}}(r) = -\sqrt{\frac{2}{\pi k}} \frac{k}{\varepsilon + 2m_e c^2} \cos(kr + \delta_\kappa), \quad (30)$$

where $\hbar k = \sqrt{E_{\text{kin}}(E_{\text{kin}} + 2m_e c^2)}$, $\varepsilon = E_{\text{kin}} + m_e c^2$, and δ_κ is the Coulomb phase shift. The energy-normalised numerical solution is then $(P_c, Q_c) = (P_{\text{num}}, Q_{\text{num}})/N$ with the normalisation factor $N = A_{\text{asympt}}/\text{amp}_{\text{rms}}$, where $A_{\text{asympt}} = \sqrt{2/(\pi k)} \sqrt{1 + E_{\text{kin}}/m_e c^2}$ and amp_{rms} is the RMS envelope amplitude $\sqrt{P^2 + Q^2}$ sampled across the converged far-region oscillation by the convergence loop of Section 3.7 (no fit of the phase is involved).

3.3 Closed-form continuum wavefunction

For a hydrogenic ion (point nucleus, $V(r) = -\alpha Z/r$) the radial Dirac equation above admits an exact solution in terms of the regular confluent hypergeometric function ${}_1F_1$, in addition to the numerical integration just described [5]. Writing the solution regular at the origin ($P, Q \sim r^s$ with $s = \sqrt{\kappa_c^2 - (z\alpha)^2}$) as

$$P_c(r) = N r^s \operatorname{Re}[u_1(r) + u_2(r)], \quad (31)$$

$$Q_c(r) = N r^s \operatorname{Re}[q_0 u_1(r) + q_2 u_2(r)], \quad (32)$$

with the ansatz

$$u_1(r) = e^{ikr} {}_1F_1(a; 2s; -2ikr), \quad a = s \left(1 + i \frac{b_1}{k}\right), \quad (33)$$

and u_2 the linear combination of u_1 and u_1' fixed by the coupled system,

$$u_2(r) = \frac{u_1'(r) - b_1 u_1(r)}{c_{12}}. \quad (34)$$

Substituting (31)–(34) into the radial system determines the real channel parameters

$$q_0 = \frac{(s + \kappa_c) c}{z}, \quad q_2 = \frac{(\kappa_c - s) c}{z}, \quad (35)$$

and

$$b_1 = \frac{E_{\text{kin}} + q_0 q_2 W_0}{c(q_2 - q_0)}, \quad c_{12} = \frac{E_{\text{kin}} + q_2^2 W_0}{c(q_2 - q_0)}, \quad (36)$$

where $W_0 = 2c^2 + E_{\text{kin}}$ (energies in Hartree, $c = 1/\alpha$) and $\hbar k = \sqrt{E_{\text{kin}}(E_{\text{kin}} + 2m_e c^2)}$ is the relativistic wavenumber of Section 3. The complex hypergeometric parameter a has $\operatorname{Re} a = s$, so both $|r^s(-2ikr)^{a-2s}|$ and $|r^s(2ikr)^{-a}|$ are $\mathcal{O}(1)$ for $r \rightarrow \infty$: the representation stays bounded in the far field.

Because ${}_1F_1$ is known, the asymptotic form of the solution is known exactly, so the energy normalisation $\int_0^\infty [P_E P_{E'} + Q_E Q_{E'}] dr = \delta(E - E')$ can be imposed directly, without the envelope-amplitude matching used by the ODE solver. The two-term Poincaré expansion of u_1 gives, for $r \rightarrow \infty$,

$$u_1(r) \sim C_{\text{in}} r^{a-2s} e^{-ikr} + C_{\text{out}} r^{-a} e^{+ikr}, \quad (37)$$

with

$$C_{\text{in}} = \frac{\Gamma(2s)}{\Gamma(a)} (-2ik)^{a-2s}, \quad C_{\text{out}} = \frac{\Gamma(2s)}{\Gamma(2s-a)} (2ik)^{-a}, \quad (38)$$

and, since $u_1' \simeq \pm ik u_1$ on the two branches, u_2 carries the branch ratios

$$t_{\text{in}} = -\frac{ik + b_1}{c_{12}}, \quad t_{\text{out}} = \frac{ik - b_1}{c_{12}}. \quad (39)$$

The far-field amplitudes of P_c and Q_c are therefore

$$C_{\text{in}}^{(P)} = (1 + t_{\text{in}}) C_{\text{in}}, \quad C_{\text{out}}^{(P)} = (1 + t_{\text{out}}) C_{\text{out}}, \quad (40)$$

$$C_{\text{in}}^{(Q)} = (q_0 + q_2 t_{\text{in}}) C_{\text{in}}, \quad C_{\text{out}}^{(Q)} = (q_0 + q_2 t_{\text{out}}) C_{\text{out}}, \quad (41)$$

and the normalisation constant is

$$N = \frac{A_{\text{asyp}}}{\sqrt{|C_{\text{in}}^{(P)}|^2 + |C_{\text{out}}^{(P)}|^2 + |C_{\text{in}}^{(Q)}|^2 + |C_{\text{out}}^{(Q)}|^2}}, \quad A_{\text{asyp}} = \sqrt{\frac{2}{\pi k}} \sqrt{1 + \frac{E_{\text{kin}}}{m_e c^2}}, \quad (42)$$

with A_{asyp} the same asymptotic amplitude used for the numerical solution above.

Numerically, ${}_1F_1(a; b; z)$ with $z = -2ikr$ is evaluated by a three-stage scheme. A Taylor series in double precision tracks the largest intermediate term; the result is trusted only while the roundoff estimate $\text{peak} \times \varepsilon_{\text{double}} / |{}_1F_1| \lesssim 10^{-8}$ ($\varepsilon_{\text{double}} = 2.2 \times 10^{-16}$), since the series converges through heavy cancellation at large $|z|$. Where that estimate fails, the same series is evaluated in quad precision (34 decimal digits, mantissa rounding 2^{-110}), which covers the moderate- $|z|$ band in which the double series has lost ~ 10 digits but the asymptotic expansion has not yet converged. For larger $|z|$ the two-term Poincaré expansion (37) is used, truncated at its smallest term (DLMF 13.7.1).

The closed form is exact in principle. Its implementation was verified pointwise against an independent arbitrary-precision (50-digit) evaluation of (31)–(42): the wavefunction (P_c, Q_c) and the energy normalisation N agree to $\sim 10^{-8}$ relative over $E_{\text{kin}} = 10^{-5}$ –1 eV for every E1-coupled κ_c (Section 3.7). Where the numerical ODE solver of the previous subsection is converged it agrees with the closed form to the ODE’s own envelope-convergence accuracy (better than 0.5% at low energy and moderate Z); at extreme low E_{kin} and high Z the ODE energy normalisation is under-converged (Section 3.7), a limitation the closed form does not share. At high E_{kin} the bound-free overlap integral (24) becomes a near-total cancellation between contributions of magnitude $\gtrsim 1$, with a residual $|R_{\text{bf}}|/S_{\text{max}} \sim 10^{-5}$ or smaller; resolving that residual to the few-percent level requires the continuum wavefunction to be accurate to $\sim 10^{-9}$ relative, beyond the $\sim 10^{-4}$ relative accuracy of the (asymptotic-truncation-limited) double-precision evaluation. The production implementation therefore uses the two solvers in a hybrid scheme keyed on the Sommerfeld parameter $\eta = Z(1 + E_{\text{kin}}/c^2)/k$: the closed form is the default for $\eta > 350$ (the low-energy/high- Z corner, where it is exact and the ODE normalisation is least reliable) and the numerical ODE approach of the previous subsection, with its Gordon fallback of Section 3, for $\eta \leq 350$ (the deep-cancellation regime). The switch and its tuning are described in Section 3.7.

3.4 Analytic (relativistic-Gordon) bound-free radial integral

The bound-free radial integral (24) can be evaluated exactly in closed form, without quadrature, by combining the closed-form continuum of Section 3.3 with the polynomial bound state of Section 1. Substituting (31)–(32) and $P_b = C\sqrt{1+\varepsilon}(2\lambda r)^\gamma e^{-\lambda r} F(r)$, $Q_b = C\sqrt{1-\varepsilon}(2\lambda r)^\gamma e^{-\lambda r} G(r)$ with $F = \sum_i a_i r^i$, $G = \sum_i b_i r^i$ into the integrand $r[P_b P_c + Q_b Q_c]$ turns every polynomial term into an integrand of the universal form

$$r^{\gamma+s+1+i} e^{-(\lambda-ik)r} {}_1F_1(a'; b'; -2ikr), \quad (43)$$

with $(a', b') = (a, 2s)$ or $(a+1, 2s+1)$. Each such term integrates exactly with the Laplace transform of ${}_1F_1$ (DLMF 13.10.7),

$$\int_0^\infty t^{\nu-1} e^{-\mu t} {}_1F_1(a'; b'; \kappa t) dt = \Gamma(\nu) \mu^{-\nu} {}_2F_1\left(a', \nu; b'; \frac{\kappa}{\mu}\right), \quad \text{Re } \nu > 0, \quad (44)$$

with $\mu = \lambda - ik$, $\kappa = -2ik$ and $\nu = \gamma + s + i + 2 > 0$. The radial integral is therefore the *finite* sum of $n_r + 1$ hypergeometric terms,

$$R_{\text{bf}} = N (2\lambda)^\gamma \text{Re} \sum_{i=0}^{n_r} \frac{\Gamma(\nu_i)}{\mu^{\nu_i}} [A_i {}_2F_1(a, \nu_i; 2s; z) + B_i {}_2F_1(a+1, \nu_i; 2s+1; z)], \quad (45)$$

with $z = -2ik/(\lambda - ik)$ and N the continuum normalisation (42). Since R_{bf} is real by construction, taking the real part outside the sum is legitimate.

The coefficients A_i, B_i assemble the branch structure of the continuum. The r -derivative of u_1 follows from the identity $\frac{d}{dz} {}_1F_1(a; b; z) = \frac{a}{b} {}_1F_1(a+1; b+1; z)$,

$$u'_1 = ik e^{ikr} \left[f(r) - \frac{a}{s} g(r) \right], \quad f = {}_1F_1(a; 2s; -2ikr), \quad g = {}_1F_1(a+1; 2s+1; -2ikr), \quad (46)$$

which with $u_2 = (u'_1 - b_1 u_1)/c_{12}$ of (34) gives the channel combinations

$$c_{f0} = \frac{ik - b_1}{c_{12}}, \quad c_{g0} = -\frac{ik a}{s c_{12}}, \quad (47)$$

$$c_{Qf} = q_0 + q_2 c_{f0}, \quad c_{Qg} = q_2 c_{g0}, \quad (48)$$

so that $u_1 + u_2 = e^{ikr}[(1 + c_{f0})f + c_{g0}g]$ and $q_0 u_1 + q_2 u_2 = e^{ikr}[c_{Qf}f + c_{Qg}g]$. Combining with the bound-state factors $\sqrt{1 \pm \varepsilon} a_i$, $\sqrt{1 \pm \varepsilon} b_i$ and the normalisation constant C of Section 1,

$$A_i = C \left[\sqrt{1 + \varepsilon} a_i (1 + c_{f0}) + \sqrt{1 - \varepsilon} b_i c_{Qf} \right], \quad (49)$$

$$B_i = C \left[\sqrt{1 + \varepsilon} a_i c_{g0} + \sqrt{1 - \varepsilon} b_i c_{Qg} \right]. \quad (50)$$

The ${}_2F_1$ values are computed recursively. The Gauss series

$${}_2F_1(a, b; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n n!} z^n, \quad (x)_n = x(x+1) \cdots (x+n-1), \quad (51)$$

is summed term by term with the recurrence $t_n = t_{n-1} (a+n-1)(b+n-1)/[(c+n-1)n] \cdot z$, each step exact up to roundoff. The series converges geometrically only for $|z| < 1$, but at high E_{kin} the argument $z = -2ik/(\lambda - ik)$ has $|z| = 2k/\sqrt{\lambda^2 + k^2} < 2$ and leaves the unit disk ($k \gg \lambda$). The portable engine therefore *recurses* through closed-form transformations to pull the argument back into the convergent region: the Pfaff transformation (DLMF 15.8.4),

$${}_2F_1(a, b; c; z) = (1-z)^{-a} {}_2F_1\left(a, c-b; c; \frac{z}{z-1}\right), \quad (52)$$

covers $|z| \lesssim 1$, and the reciprocal connection formula (DLMF 15.8.2),

$${}_2F_1(a, b; c; z) = A_1(-z)^{-a} {}_2F_1\left(a, a+1-c; a+1-b; \frac{1}{z}\right) + A_2(-z)^{-b} {}_2F_1\left(b, b+1-c; b+1-a; \frac{1}{z}\right), \quad (53)$$

with A_1, A_2 ratios of gamma functions, covers $|z| > 1$. Each transformed argument is evaluated by re-entering the same summing routine, hence the recursion; the depth is bounded so the chain always terminates, and a stage is accepted only when its roundoff estimate $\text{peak} \times \varepsilon / |{}_2F_1| \lesssim 10^{-8}$ holds (double precision, with a quad-precision re-summation in the band around $|z| = 1$). In builds with FLINT the exact ball-arithmetic engine `acb_hypgeom_2f1` evaluates all the ${}_2F_1$'s by automatic analytic continuation instead. If no stage is trustworthy (portable engine only), the analytic element returns failure and the caller falls back to the numerical path.

The analytic element is the relativistic analog of the nonrelativistic Gordon element: it is exact at every E_{kin} , so it is free of the deep-cancellation loss that limits the numerical quadrature at high energy. Within the dipole path of `dirac_rr_xsec()` it is the default whenever the code is built with FLINT (the fully-retarded path of Section 3.5 takes precedence when available); `HYDROCAL_DIRAC_GORDON` selects the dipole radial-integral path explicitly (0 = legacy numerical path of the next subsection, 1 = analytic path). It was verified against an independent 50-digit (mpmath) evaluation of (45) on the `Ds3p32DiracRR` $Z = 110$ scan: agreement is within $\sim 3 \times 10^{-11}$ relative at every energy $E_{\text{kin}} = 10^{-5} - 10^3$ eV for all three E1-coupled channels $\kappa_c = -1, -3, +2$, including the deep-cancellation high-energy points, and the identity (44) was confirmed against direct quadrature of (43) to 10^{-19} relative.

3.5 Fully-retarded (plane-wave) bound-free matrix element

`dirac_rr_xsec()` closes the dipole gap discussed at the start of this section with the fully-retarded plane-wave matrix element of Section 2.2, applied to the bound-free case. The retarded

operator $e^{-i\mathbf{k}_p \cdot \mathbf{r}}$ (photon wavenumber $k_p = \omega/c$, with ω in Hartree) replaces the dipole r factor; expanding the plane wave in spherical partial waves

$$e^{-i\mathbf{k}_p \cdot \mathbf{r}} = \sum_L (2L+1)(-i)^L j_L(k_p r) P_L(\cos \theta), \quad (54)$$

the channel radial integrals are the cross-type elements

$$R_+[L] = \int_0^\infty P_c(r) Q_b(r) j_L(k_p r) dr, \quad (55)$$

$$R_-[L] = \int_0^\infty Q_c(r) P_b(r) j_L(k_p r) dr, \quad (56)$$

with (P_c, Q_c) the continuum and (P_b, Q_b) the bound state. Both parities and all multipole orders are allowed — the angular coefficients of `ang_int()` decide — so the continuum channel set $|j_c - j_b| = 0, 1, 2, \dots$ is complete, each j_c realised by both $\kappa_c = \pm(j_c + \frac{1}{2})$; the largest multipole order of a channel is $L_{\max} = \max[l(\kappa_c) + l(-\kappa), l(-\kappa_c) + l(\kappa)]$, exactly as in Section 2.2. The per-channel matrix-element sum

$$tot(\kappa_c) = \sum_{m_c, m_b, \hat{\epsilon}} |M|^2, \quad (57)$$

is assembled precisely as the bound-bound `retarded_total()` of Section 2.2: the angular factor of each pair of spinor harmonics is reduced by `ang_int()` to coefficients of $j_L(k_p r)$, and the two photon polarisations are combined with the M_x, M_y phase conventions of that section ($M_x = i[(A_0 B_3) + (A_1 B_2) - (A_2 B_1) - (A_3 B_0)]$, $M_y = (A_0 B_3) - (A_1 B_2) - (A_2 B_1) + (A_3 B_0)$, with $+i$ for large $\times$ small and $-i$ for small $\times$ large pairs).

The radial integrals (56) are evaluated by three routes. (i) *Closed form*. With the closed-form continuum of Section 3.3 and the polynomial bound state of Section 1, the decomposition

$$j_L(k_p r) = \frac{1}{2} \sum_{m=0}^L C_{Lm}(k_p r)^{-(m+1)} [i^{m-L-1} e^{+ik_p r} + (-i)^{m-L-1} e^{-ik_p r}], \quad C_{Lm} = \frac{(L+m)!}{2^m m! (L-m)!}, \quad (58)$$

turns every polynomial term of the integrand into the universal form $r^{\gamma+s+i-m-1} e^{-\mu r} {}_1F_1(a'; b'; -2ik_c r)$ with $\nu = \gamma + s + i - m$ and the two branches $\mu_{\pm} = \lambda - i(k_c \pm k_p)$ (the $e^{\pm ik_p r}$ branches), which integrates exactly via the Laplace transform of ${}_1F_1$ (DLMF 13.10.7, Eq. 44) to $\Gamma(\nu) \mu^{-\nu} {}_2F_1(a', \nu; b'; -2ik_c/\mu)$. $R_{\pm}[L]$ are therefore finite sums of $\Gamma(\nu) \mu^{-\nu} {}_2F_1(\dots)$ values — the retarded generalisation of the dipole analytic element of Section 3.4, exact at every energy, all multipoles, no quadrature. The alternating m -sum is accumulated with Kahan compensation and screened: a multipole whose cancellation depth $\text{peak}/|\text{sum}|$ exceeds 10^{12} , or that lies beyond the bound-state turning point $L > k_p r_{\text{bound}}$ with $r_{\text{bound}} = 2(n_r + \gamma)/\lambda$ and shows any residual cancellation, is flagged unreliable and recomputed by route (iii). (ii) *Small- k_p Taylor form*. Below $k_p \approx 0.5$ the branch terms of order $k_p^{-(m+1)}$ cancel to leave $j_L \sim (k_p r)^L$ at a depth beyond double precision ($Z = 1$, $E_{\text{kin}} = 1$ eV, $k_p \sim 0.004$: depth $\sim 10^{229}$), and $\Gamma(\nu)$ approaches a pole as $\nu = \gamma + s + i - m \rightarrow 0$ (i.e. $m \sim s + i$ at low Z); the whole channel is then evaluated with the convergent Taylor series

$$j_L(x) = \sum_t (-1)^t \frac{x^{L+2t}}{2^t t! (2L+2t+1)!!}, \quad (59)$$

and the power-law integrals $I_{\pm}[p] = \int_0^\infty r^p P_c Q_b dr$, $I_{-}[p] = \int_0^\infty r^p Q_c P_b dr$ in closed form via the same Laplace-transform trick with the always-positive exponent $\nu = \gamma + s + i + p + 1$ (never a pole, no $k_p^{-(m+1)}$ factors):

$$R_{\pm}[L] = \sum_t (-1)^t \frac{k_p^{L+2t}}{2^t t! (2L+2t+1)!!} I_{\pm}[L+2t], \quad (60)$$

cached per exponent $p = L + 2t$; the route fails (and the caller falls back) if the t -series does not converge within $t_{\max} = 40$ terms or suffers deep cancellation. (iii) *Grid quadrature*. Flagged multipoles — or the whole channel, when the closed-form ${}_2F_1$ evaluation was not trustworthy (portable engine without FLINT) — are computed by trapezoidal quadrature on the radial grid with the ODE continuum of Section 3 and pointwise j_L . The direct quadrature has no branch conditioning and reproduces the validated channel totals to well below 1%.

The fully-retarded cross section per vacancy follows by summing the channel contributions over j_c :

$$\sigma_{\text{RR}}^{\text{ret}}(n, \kappa; E_{\text{kin}}) = \frac{2\pi^2}{\alpha\omega} \frac{\sum_c \text{tot}(\kappa_c)}{2j_b + 1} \frac{\alpha^2 \omega^2}{2k_{\text{el}}^2} a_0^2, \quad (61)$$

the fully-retarded analogue of the long-wavelength Milne formula (22); as $k_p \rightarrow 0$ it reduces to the dipole result. The per-channel contributions are positive definite, so the j_c shells are added in ascending order until one shell adds less than 10^{-4} of the running total (the series is capped at $|j_c - j_b| = 80$); the independent per-channel closed-form evaluations are parallelised (OpenMP) over the shells.

The retarded path is the default in `dirac_rr_xsec()` when the exact ${}_2F_1$ engine (FLINT) is available, since the closed-form retarded integrals need it; without FLINT the dipole path of Section 3.4 is the default. `HYDROCAL_DIRAC_RETARDED=0` forces the dipole path (validation/legacy), any other non-empty value forces the retarded path, and `HYDROCAL_DIRAC_RETARDED_NUMERIC=1` forces the grid quadrature of route (iii) for every channel (validation cross-check).

3.6 Radial integral and cross section

In the dipole path the bound-free radial integral (24) is evaluated by the analytic closed form of Section 3.4, which is the default of that path when the code is built with FLINT. The legacy numerical path of this section — the trapezoidal integration on the common logarithmic grid used for the continuum integration, with the bound-state radial functions (P_b, Q_b) evaluated analytically at each grid point via the polynomial expansion of Section 1, and with the nonrelativistic analytic Gordon element substituted (via a runtime-calibrated conversion factor) wherever the oscillatory cancellation of the integral exceeds what double precision can sustain at high E_{kin} and high ℓ (see Section 3.7, *High-energy ℓ channels*) — is retained as the fallback for builds without FLINT and as a cross-check. Finally, the RR cross section is obtained from (22). In the non-relativistic limit it reduces to the familiar form

$$\sigma_{\text{RR}}^{\text{nr}}(n, l; E_{\text{kin}}) = \frac{4\pi\alpha a_0^2}{3} \frac{2l + 1}{2} \frac{1}{E_{\text{kin}}} |\langle l \| C^{(1)} \| l_c \rangle|^2 |R_{\text{bf}}^{\text{nr}}|^2, \quad (62)$$

which reproduces the Kramer–Kramers formula after summing over l .

3.7 Validation against Ichihara & Eichler (2000)

The point-nucleus $Z = 1$ cross sections of `dirac_rr_xsec()` were compared against the “exact relativistic” tabulated cross sections of Ichihara and Eichler [6] for capture into the $1s_{1/2}$ – $3d_{5/2}$ states, over the full set of 38 electron kinetic energies (1 eV–50 keV) printed for $Z = 1$ in that table. Agreement with the $Z = 1$ table is within a few percent or less across the tabulated energy range.

High- ℓ capture channels: near-origin under-resolution. The comparisons above exercise only capture into $1s$ – $3d$ bound states, whose E1-coupled continuum partial waves have $|\kappa_c| \leq 4$ (Section 3); they therefore never exercise the channels that a second, independent defect destroys. Comparing `dirac_rr_xsec()` instead against the non-relativistic Kramers-type formula `sigmarrqm()` of the non-relativistic code (RRDRxsec.cxx) at $Z = 1$, $E_{\text{kin}} = 0.1$ eV showed agreement with the physically sensible QMD values to $\sim 0.3\%$ for $l \leq 3$, but catastrophic disagreement

for $l \geq 4$: the DIR cross sections were $\sim 40\times$ too small at $l = 4$ and $10^6\text{--}10^{16}\times$ too small for $l = 5\text{--}7$, the error growing with both l and n .

Root cause (high ℓ). The radial ODE is integrated outward from $r_0 = 10^{-6}$, where the true solution scales as $P, Q \sim r^{\gamma_c}$ with $\gamma_c = \sqrt{\kappa_c^2 - (Z\alpha)^2}$. For the continuum channels coupled to high- ℓ bound states, $|\kappa_c|$ is large and $\gamma_c \simeq |\kappa_c|$ (e.g. $\gamma_c \approx 6$ for $|\kappa_c| = 6$ at $Z = 1$), so the starting values are $P_0, Q_0 \sim 10^{-36}$ — six orders of magnitude below the stepper’s absolute error floor of 10^{-30} . The error controller there treats the near-origin transient as numerical noise: the first, unavoidably coarse steps seed a spurious mode that persists to the far field and corrupts both the amplitude and the phase of the energy-normalised continuum wavefunction. The defect is in the *representation*, not in the convergence of the envelope loop: tightening that loop’s tolerance a further $50\times$ made the high- ℓ ratios *worse* (down to $\sim 0.2\text{--}0.24$ instead of 0.9974), and a purely rescaled integration fixed $l \leq n - 2$ but left a residual drift at $l = n - 1$ — in rescaled variables $p, q \sim A/r^\gamma$ the far-field state itself collapses back below the absolute floor, so the raw variables are the correct far-field representation.

Fix (high ℓ). `integrate_PQ()` now integrates the rescaled variables $p = P/r^{\gamma_c}$, $q = Q/r^{\gamma_c}$ — $\mathcal{O}(1)$ at r_0 for every κ_c , so the absolute tolerance acts on a well-conditioned state — while the raw state is small, and switches to the raw variables once $|P|, |Q| > 10^{-25}$, beyond which the relative tolerance governs. With this change the agreement becomes uniform: $QMD/DIR = 0.9974$ for *every* ℓ at $Z = 1$, $E_{\text{kin}} = 0.1\text{ eV}$, including the previously worst case $\ell = n - 1$ (e.g. $n = 8$, $\ell = 7$); spot checks at $Z = 8$, $E_{\text{kin}} = 0.1\text{ eV}$ and $Z = 1$, $E_{\text{kin}} = 10\text{ eV}$ are likewise uniform across all ℓ (~ 0.991 and ~ 0.9995 , respectively), with no residual ℓ -dependence. The uniform $\sim 0.3\%$ offset from unity at $Z = 1$, $E_{\text{kin}} = 0.1\text{ eV}$ reflects the relativistic treatment of `dirac_rr_xsec()`, which the non-relativistic Kramers formula does not include; it shrinks as E_{kin} rises ($\sim 0.05\%$ at 10 eV). The envelope-convergence parameters themselves were left at the values described in the *Fix* paragraph above.

High-energy ℓ channels: catastrophic cancellation and the Gordon fallback. The validations above exercise only $E_{\text{kin}} \lesssim 100\text{ eV}$ at $Z = 1$ (or the most compact states at higher Z), and so never expose a third, independent defect of the bound-free radial integral. Comparing `dirac_rr_xsec()` against the non-relativistic Kramers-type formula `sigmarrqm()` at $Z = 1$, $E_{\text{kin}} \geq 100\text{ eV}$ showed smooth, uniform agreement at low ℓ ($QMD/DIR \simeq 1.0045$ at 1 keV and $\simeq 1.035\text{--}1.046$ at 10 keV , growing with E_{kin} as the relativistic treatment departs from the Kramers limit), but wildly inconsistent values for high- ℓ channels: at 100 eV , capture into $n = 8$, $\ell = 7$ gave $QMD/DIR \simeq 1.12$, and at $1\text{--}10\text{ keV}$ the high- ℓ ratios ranged from 0.0000 to $\sim 10^9$, with no smooth energy or ℓ trend.

Root cause (high energy). The bound-free integral (24) is oscillatory: the continuum partial wave oscillates over its full energy range, and the trapezoidal scan accumulates a running sum whose peak magnitude S_{max} is of order $R_{\text{bf}} \times (\text{number of periods})$ — enormously larger than the final, physically meaningful value, which emerges only through near-total cancellation of equal positive and negative lobes. At $Z = 1$, $E_{\text{kin}} = 10\text{ keV}$, $n = 8$, $\kappa = -8$ the partial sums build to $|S_{\text{max}}| \sim 0.05$ while the true integral is $R_{\text{bf}} \sim 2 \times 10^{-22}$: the cancellation depth is $\sim 5 \times 10^{20}$, far beyond the $\sim 10^{16}$ that double precision can sustain, and the residual trapezoidal value is dominated by accumulated roundoff. This is a *representation* defect, not a convergence one: refining the radial grid, tightening the ODE tolerance, or accumulating in long double all leave the error essentially unchanged (indeed an $8\times$ grid refinement at 10 keV made the high- ℓ answer *worse*).

Fix (high energy). The non-relativistic bound-free radial matrix element has an exact closed form — the Gordon (Infeld–Hull) recursion, already implemented in the code as `CalcRbc()` in `osci.cxx`. For $Z = 1$ the relativistic correction to the radial element itself is only $O((Z\alpha)^2) \simeq 5 \times 10^{-5}$, so substituting the analytic element is asymptotically correct in exactly the regime where the numerics fail. In `dirac_rr_xsec()` each channel’s trapezoidal R_{bf} is kept only if $|R_{\text{bf}}| > 10^{-8} S_{\text{max}}$, i.e. the cancellation depth leaves it at least eight orders of magnitude above the roundoff floor; otherwise the analytic Gordon element is substituted via

$$R_{\text{bf}} = c \frac{2}{z^2} R_{\text{Gordon}} \left(\frac{1}{\sqrt{E_{\text{kin}}/(z^2 \text{ Ryd})}} \right), \quad (63)$$

using the $l \rightarrow l + 1$ column `Rbcm[l+1]` for $\Delta l = +1$ channels and the $l \rightarrow l - 1$ column `Rbcp[l]` for $\Delta l = -1$ channels (mapping verified against the numerical integration to $\sim 0.05\%$ where the latter is reliable). The residual factor c is calibrated at runtime from the *reliable* channels of the same (E_{kin}, Z) — $c = \langle R_{\text{bf}}^{\text{num}} / ((2/z^2) R_{\text{Gordon}}) \rangle$ averaged over channels passing the reliability test — because c carries the relativistic continuum normalisation and drifts from ~ 1.0003 at low energy to ~ 0.978 at 10 keV for $Z = 1$. The substitution is applied only where it is a legitimate asymptotic approximation: $(Z\alpha)^2 < 0.1$ (i.e. $Z \lesssim 43$; a closed form exists for the hydrogenic *continuum wavefunction* itself, Section 3.3, but not for the bound-free overlap integral of two Dirac radial functions), and the calibrated c must be consistent across the reliable channels (relative scatter $< 20\%$, computed from the running mean and second moment of c). When the guard fails, the unreliable channel is *dropped* (it contributes exactly zero) rather than assigned a wrong value. Finally, the R_{bf}^2 of the highest- ℓ high-energy channels falls below the old `total_me_sq` $< 10^{-30}$ early-return floor; that floor was lowered to 10^{-100} (safe, because every contribution is a non-negative square of a reliable-or-analytic element).

Result (high energy). The agreement is now uniform across all ℓ at every energy. At $Z = 1$: $QMD/DIR \simeq 1.0001$ – 1.0007 at 100 eV, $\simeq 1.0017$ – 1.0047 at 1 keV, and $\simeq 1.0346$ – 1.0455 at 10 keV, with no residual 0.0000 or 10^9 rows; the previously failing $n = 8, \ell = 7$ channel went from 0.0000 to 1.0017 at 1 keV. Spot checks at $Z = 2, 1$ keV are likewise uniform (~ 1.0007 – 1.006). The small spread within each energy — high- ℓ ratios running slightly *below* the $\ell = 0$ value (e.g. 1.0348 vs 1.0455 at 10 keV) — is the known limitation of substituting a single calibrated scalar c for all channels: the true per-channel factor has a weak ℓ - and j -dependence (the two κ_c sub-channels of one transition differ by $\sim 0.3\%$ at 1 keV and $\sim 3\%$ at 10 keV for $Z = 1$, a genuine relativistic splitting that a scalar cannot reproduce). The low-energy behaviour documented above (0.9974/0.9995/0.991, $Z = 1/Z = 1/Z = 8$ at 0.1/10/0.1 eV) is unchanged.

Validity regime of the fallback (high Z). The substitution is only as accurate as the non-relativistic element it inserts, so its asymptotic validity is set by the smallness of $(Z\alpha)^2$ (5×10^{-5} at $Z = 1$, $\sim 45\%$ at $Z = 92$); the guard of the previous paragraph is what enforces this regime. At $Z = 92$ the fallback is inert up to at least 100 keV: there is no catastrophic cancellation there (all 43 channels of the 100 keV scan pass the reliability test), so the numerical integrals are kept unchanged. At 1 MeV the cancellation does set in for $\ell \geq 5$, but the per-channel relativistic restructuring is so strong that no single scalar c can represent it — the calibrated mean is $c \simeq 1.109$ while the individual reliable channels scatter from -0.50 to $+4.35$, far beyond the 20% guard, and substituting would both err by $O((Z\alpha)^2) \simeq 45\%$ and erase the genuine ℓ, j dependence (the pre-guard code collapsed all of $\ell = 5, 6, 7$ onto one identical ratio). The guard therefore drops those channels rather than substitutes; the resulting undercount is preferable to silently wrong physics. In short, the Gordon fallback is validated only for $Z \lesssim 43$; for higher Z the comparison to `sigmarrqm()` is not meaningful above $E_{\text{kin}} \sim$ a few hundred keV, where the code was never validated against data. In builds with FLINT the fully relativistic analytic

element of Section 3.4 replaces this calibrated fallback entirely: it is exact at every Z and E_{kin} , so none of the $(Z\alpha)^2$, calibration-scatter, or $Z \lesssim 43$ restrictions of this paragraph apply.

Extended validation, $Z = 1$ –112. The comparison was extended to higher Z using the per- Z tables printed by [6] (its Tables for $Z = 1$ through $Z = 112$; the $Z = 61$ and $Z = 70$ tables could not be digitized because of a text-layer defect specific to those two pages in the source PDF), drawing a representative sample of Z values and energies from the digitized data rather than an exhaustive point-by-point sweep of all 112 tables. Table 3 summarises the ratio $\sigma_{\text{RR}}^{\text{dirac_rr_xsec}}/\sigma_{\text{RR}}^{\text{Ichihara}}$ for the $1s_{1/2}$ state over $E_{\text{kin}} = 1\text{ eV}$ – 10 keV , at a representative spread of Z , before and after the fix above (before: using the original fixed-period, single-comparison convergence loop).

Z	median ratio (before)	median ratio (after)	mean abs. dev. (after)
1	0.94–1.08	0.997	1.0%
30	~ 1.30	0.998	1.1%
50	~ 1.52	1.005	1.3%
90	~ 1.7 – $2.1\times$	1.038	3.3%
110	~ 1.6 – $2\times$ (worst points off by $\gtrsim 2\times$)	1.100	9.2%
112	worst points off by up to $\sim 2.3\times$	1.109	10.1%

Table 3: Ratio of `dirac_rr_xsec()` to the tabulated Ichihara & Eichler $1s_{1/2}$ cross section, before and after the convergence-loop fix, at representative Z (5 energies per Z , 1 eV–10 keV). “Before” figures are approximate, summarising the spread seen for the uncorrected code; “after” figures are the median ratio and mean absolute fractional deviation over the same energy set, using the final convergence tolerance described above.

The fix removes the dominant, rapidly- Z -growing error (up to $\sim 130\%$ at $Z = 112$ under the old code) and brings agreement back to the few-percent level found for $Z = 1$, generalising cleanly across other tabulated states ($2s$ – $3d$), not only $1s_{1/2}$. A smaller residual trend remains, growing smoothly (no discontinuity or sudden onset) from $\sim 1\%$ at $Z = 1$ to ~ 9 – 10% at $Z \gtrsim 110$, concentrated at the lowest tabulated energies.

Closed-form continuum cross-check. `dirac_rr_xsec()` selects its continuum solver by the Sommerfeld parameter $\eta = Z(1 + E_{\text{kin}}/c^2)/k$: by default (no environment variable set) the exact ${}_1F_1$ construction of Section 3.3 is used for $\eta > 350$ and the numerical ODE integration of Section 3 for $\eta \leq 350$. Setting `HYDROCAL_DIRAC_CLOSED=0` forces the ODE everywhere (the legacy behaviour); any other non-empty value forces the closed form everywhere (the validation cross-check; it is slower per grid point, and the exact ${}_1F_1$ engine is used only when built with FLINT).

Two checks were made with the closed form on the $Z = 110$, capture-to- $3p_{3/2}$ scan of the `Ds3p32DiracRR` example ($E_{\text{kin}} = 10^{-5}$ – 10^3 eV). First, the closed-form wavefunction (P_c, Q_c) and its energy normalisation N were compared pointwise against an independent 50-digit (mpmath) evaluation of (31)–(42); they agree to $\sim 10^{-8}$ relative at both $E_{\text{kin}} = 10^{-5}$ and 1 eV for all three E1-coupled continuum channels $\kappa_c = -1, -3, +2$, across the entire radial range of the overlap integral — the residual is set by the double/quad-precision ${}_1F_1$ evaluation, not by the formula. Second, the cross sections were compared against the ODE path. The ratio $\sigma_{\text{closed}}/\sigma_{\text{ODE}}$ is 0.455 at $E_{\text{kin}} = 10^{-5}$ eV and rises monotonically through 0.983 at 1 eV to 0.997 at 10^3 eV. The closed-form and ODE continuum wavefunctions have *identical* r -dependence (the two are proportional at every radius), so the cross-section gap is a pure normalisation effect: at $E_{\text{kin}} = 10^{-5}$ eV the ODE envelope-matched solution is ~ 1.4 – $1.5\times$ the exact (closed-form) one depending on the channel, $1.009\times$ at 1 eV, and $1.001\times$ at 10^3 eV. Since the closed form is the exact solution, it is the ODE energy normalisation of Section 3 that is in error at this extreme Sommerfeld

parameter $\eta = Z(1 + E_{\text{kin}}/c^2)/k$: its envelope-convergence loop stops while the RMS amplitude is still slowly drifting (the same limitation already documented above at $Z = 112$), and the gap closes monotonically as E_{kin} rises and η shrinks. The low-energy closed-form R_{bf} passes the $|R_{\text{bf}}| > 10^{-8} S_{\text{max}}$ reliability test, so the discrepancy is not the Gordon-fallback logic; it is the ODE normalisation. The closed form therefore provides an exact reference precisely in the low-energy-high- Z corner where the ODE envelope loop is least reliable, and the two paths agree everywhere the ODE is converged — which is why the default hybrid scheme of the previous paragraph (closed form for $\eta > 350$) puts the exact solution exactly where the numerical solver is weakest. In a build with FLINT this whole comparison is moot in the default configuration: the bound-free element is computed by the closed forms of Section 3.4 (dipole path) or Section 3.5 (fully-retarded path) without any continuum wavefunction, so the normalisation-gap analysis above applies only to the legacy numerical path.

Fully-retarded cross section. The fully-retarded plane-wave path of Section 3.5 is the physically complete all-multipole bound-free calculation, and it is what `dirac_rr_xsec()` uses by default. The Ichihara & Eichler tables [6] are themselves “exact relativistic” — they include all multipole orders as well as finite nuclear size — so the fully-retarded path reproduces them to the level of the point-nucleus idealisation, including the high-energy regime in which the dipole path described above falls short by orders of magnitude (e.g. a factor ~ 500 at $Z = 110$, $n = 3$, 2 MeV). As a validation cross-check, `HYDROCAL_DIRAC_RETARDED_NUMERIC=1` forces the grid quadrature (route iii of Section 3.5) for every channel; the direct quadrature has no branch conditioning at all and reproduces the closed-form channel totals to well below 1%.

Grid-quadrature continuum: normalisation bias below 100 eV. The fully-retarded path also evaluates the radial integrals on a grid whenever the closed-form m -branch sum flags a multipole (route iii of Section 3.5); at $Z = 92$ that happens for the majority of multipoles of *every* channel, so the grid quadrature is in practice the production engine there. Until now that quadrature always took its continuum from the ODE solver of Section 3, and the two continua differ only by a constant (their r -dependence is identical), so the quadrature inherited precisely the energy-normalisation bias quantified in the previous paragraph. Measured for capture into $3d_{5/2}$ at $Z = 92$ (point-nucleus DIR against the table): $\sigma_{\text{DIR}}/\sigma_{\text{IE}} = 1.0154$ at 1.5 eV, 1.0089 at 10 eV, 1.0057 at 50 eV, 1.0037 at 100 eV, 1.0026 at 10^3 eV — a smooth threshold-like excess equal to twice the measured normalisation error ($+0.72\%$ in (P_e, Q_e) at 1.5 eV $\Rightarrow +1.44\%$ in σ , $+0.24\% \Rightarrow +0.47\%$ at 100 eV, and so on). Taking the continuum from the exact ${}_1F_1$ construction (`dirac_continuum_PQ_closed`) inside the grid quadrature whenever FLINT is available removes the bias entirely: the same 14 energies from 1.5 eV to 10 MeV then give $\sigma_{\text{DIR}}/\sigma_{\text{IE}}$ between 0.9980 and 1.0022, flat at the three-significant-digit rounding width of the printed table (cost a factor ~ 1.5 –2 per energy point; the regression tests are unaffected). This also corrects the earlier reading of the residual as a point-nucleus versus finite-nuclear-size effect: for $3d_{5/2}$ that effect is below the precision of the table once the normalisation bias is gone. In a build without FLINT the exact engine is unavailable, so a residual $+0.2$ – 1.4% excess remains below 1 keV and is the documented accuracy limit of the no-FLINT fully-retarded cross sections at high Z .

4 Tabulated Ichihara & Eichler cross sections by interpolation

For applications that need the literature values themselves, rather than the Dirac-equation calculation of Section 3, `sigma_IchiharaEichlerRR()` (declared in `DiracRR.h`) returns $\sigma_{\text{RR}}(Z, n, l, j; E_{\text{kin}})$ (cm^2) at an arbitrary electron kinetic energy by interpolating the digitized tables of [6] for $Z = 1$ –112 and the nine tabulated states $1s_{1/2}$, $2s_{1/2}$, $2p_{1/2}$, $2p_{3/2}$, $3s_{1/2}$, $3p_{1/2}$, $3p_{3/2}$, $3d_{3/2}$, $3d_{5/2}$ (the digitized values themselves live in `IchiharaEichlerRRdata.h`).

Since the tabulated cross sections span many orders of magnitude and are close to power-law in E_{kin} , each (Z, state) pair's data are interpolated with a natural cubic spline (`spline()/splint()`, `hydromath.h`), in $\log_{10} E_{\text{kin}} - \log_{10}(\sigma E_{\text{kin}})$ space rather than $\log_{10} \sigma$ directly: since $\sigma_{\text{RR}} \sim 1/E_{\text{kin}}$ near threshold, σE_{kin} is a smoother interpolant than σ itself. Because the printed table stops listing digits for a state once σ becomes too small for the page format, each state's spline is built only over its own sub-range of the tabulated energy grid for that Z , not the full range. `sigma_IchiharaEichlerRR()` returns exactly 0 for an energy outside a given state's own tabulated range, or for a (Z, n, l, j) that is not one of the nine tabulated states or outside $Z = 1\text{--}112$, rather than clamping to the nearest tabulated point: clamping was found to overestimate the (rapidly falling) true cross section by more than an order of magnitude when queried far beyond a state's tabulated range.

5 Data flow in the cascade calculation

The cascade calculation in `RRphotons.cxx` treats the population of excited states as a quasi-stationary linear system. The code path for $Z > 50$ (or when the environment variable `HYDROCAL_NREL` is set) uses a mixed basis: states with $n \leq N_{\text{rel}}$ are treated relativistically (Dirac spinors), while higher- n states use the non-relativistic hydrogenic approximation.

5.1 State ordering

The basis states are a mixture of relativistic Dirac states (with definite κ , i.e. definite j) and non-relativistic hydrogenic states (with orbital l only). All states are sorted by increasing binding energy (the ground state has index 0). Relativistic states within a given n shell are ordered by $|\kappa|$ (which correlates with energy), so that higher-energy states always have larger indices. This guarantees that all transitions flow from higher to lower indices, simplifying the cascade matrix to lower-triangular form.

5.2 Rate matrix

In the relativistic part every state has a well-defined total angular momentum j (encoded in κ). All transitions are therefore j -resolved: a rate $A(n, \kappa \rightarrow n', \kappa')$ connects individual fine-structure levels, and the photon energy $\omega = |E_{n\kappa} - E_{n'\kappa'}|$ depends on both j values. This differs from the non-relativistic treatment, where rates are l -averaged and later partitioned by statistical weight.

For each upper state i (in order of increasing energy), the diagonal of the rate matrix is

$$M_{ii} = -1/\tau_i,$$

where τ_i is the total lifetime summing E1, E2, and M1 decays to all lower j -resolved states. For relativistic lower states j ,

$$M_{ji} += A_{\text{total}}(i \rightarrow j),$$

so that E1, E2, and M1 transitions each enter as individual matrix entries and feed the population of the lower fine-structure level. This is consistent with the emitted cascade lines, which are built from the same $A_{\text{total}}(i \rightarrow j)$, so that no decay channel is missing in the population redistribution. For non-relativistic upper states decaying to relativistic fine-structure components, the non-relativistic E1 rate is partitioned by statistical weight:

$$M_{ji} += A_{\text{E1}}^{\text{nr}}(n_i, l_i \rightarrow n_j, l_j) \frac{2j_j + 1}{2(2l_j + 1)}.$$

The ground state (index 0) is excluded from the solved system because it has no radiative decay. Removing its row and column gives the reduced matrix M_{exc} of size $(N_{\text{states}} - 1) \times (N_{\text{states}} - 1)$.

5.3 Population solution

The right-hand side vector is the state-resolved, fine-structure *level-total* radiative recombination cross section,

$$rhs_{i-1} = -\sigma_{RR}(i), \quad i = 1, \dots, N_{\text{states}} - 1.$$

For relativistic states,

$$\sigma_{RR}(i) = (2j_i + 1) \sigma_{RR}^{\text{vac}}(n_i, \kappa_i; E_{\text{kin}}), \quad (64)$$

where σ_{RR}^{vac} is the per-vacancy cross section returned by `dirac_rr_xsec()` (Eq. 22, obtained from the Milne relation via a numerical continuum integration and bound-free radial integral, Section 3); the explicit $(2j_i + 1)$ factor sums the per-vacancy cross section over the fine-structure level's magnetic substates to give the level-total capture cross section, consistent with the level-total lifetimes τ_i and rates $A(i \rightarrow j)$ used elsewhere in M . For non-relativistic states, `sigmarrqm()` (the Kramers–Kramers formula) already returns the full nl -subshell total directly, with no extra factor needed; it is partitioned among fine-structure components by statistical weight when the destination is relativistic, exactly as in the rate matrix (Section 5.2). The quasi-stationary populations x_i (populations per unit electron flux) satisfy

$$M_{\text{exc}} \mathbf{x} = \mathbf{rhs},$$

solved by LU decomposition. The resulting populations enter the spectrum building as described below.

5.4 Construction of the photon spectrum

Each recombination event produces either a direct RR photon or a cascade photon.

Direct RR lines

For every state i (including the ground state), a line at energy

$$E_{\text{photon}} = E_{\text{kin}} - E_{\text{bind}}(i)$$

is recorded with strength $\sigma_{RR}(i)$ (zero width), the same level-total cross section defined in Section 5.3 (Eq. 64 for relativistic states, already including the $(2j + 1)$ degeneracy factor).

Cascade lines

For each excited state $i > 0$ with population x_{i-1} and each lower state $j < i$, the cascade line parameters are

$$\text{strength} = A(i \rightarrow j) \times x_{i-1}, \quad (65)$$

$$\text{energy} = E_{\text{bind}}(i) - E_{\text{bind}}(j), \quad (66)$$

$$\text{width} = \hbar (1/\tau_i + 1/\tau_j), \quad (67)$$

where the transition rate $A(i \rightarrow j)$ uses the appropriate function depending on the type of the two states:

$$A(i \rightarrow j) = \begin{cases} A_{\text{total}}(n_i, \kappa_i \rightarrow n_j, \kappa_j) & \text{relativistic} \rightarrow \text{relativistic} \\ A_{\text{E1}}^{\text{nr}}(n_i, l_i \rightarrow n_j, l_j) \frac{2j_j + 1}{2(2l_j + 1)} & \text{non-rel.} \rightarrow \text{relativistic} \\ A_{\text{E1}}^{\text{nr}}(n_i, l_i \rightarrow n_j, l_j) & \text{non-rel.} \rightarrow \text{non-relativistic} \end{cases}$$

In the relativistic case the rate and the binding energies are j -resolved: each state is labelled by (n, κ) and the photon energy $\omega = |E_{n_i \kappa_i} - E_{n_j \kappa_j}|$ depends on both fine-structure components. For

non-relativistic upper states decaying to relativistic lower states, the l -averaged rate is distributed among the fine-structure levels by their statistical weights, ensuring that every line in the final spectrum connects states with definite j wherever the lower state is relativistic.

Each ordered pair (i, j) of upper and lower state produces exactly one line in the output. When the E1 channel is forbidden between the two states, A_{total} returns either the M1 rate, the E2 rate, or their sum, and the line is recorded with that rate and the corresponding j -resolved photon energy. Hence, M1 and E2 transitions are not averaged or lumped with other decays but appear as individually resolved spectral lines at their characteristic fine-structure energies. This is essential for correctly predicting, for example, the $2p_{3/2} \rightarrow 2p_{1/2}$ M1 line in highly charged ions.

The natural line width is the sum of the decay widths of upper and lower state (the lower-state contribution is zero for the $1s$ ground state).

5.5 Output

The lines are sorted by energy and written to a `.RRlines` file with columns: energy (eV), cross-section (cm^2), width (eV), and a transition label.

6 Implementation notes

All bound-bound rate functions, the combined dispatch, and the lifetime are declared in `DiracRate.h` and implemented in `DiracRate.cxx`. The RR cross section and the Ichihara & Eichler interpolation are declared in `DiracRR.h` and implemented in `DiracRR.cxx` (the digitized Ichihara & Eichler data itself lives in `IchiharaEichlerRRdata.h`). The E1, E2, and (different- n) M1 rates are all evaluated with the fully retarded plane-wave matrix element of Section 2.2 (`retarded_total()`), which avoids the normalisation and selection-rule bugs of the former long-wavelength multipole formulas. The RR cross section `dirac_rr_xsec()` uses by default the fully-retarded plane-wave bound-free matrix element of Section 3.5 (`dirac_rr_xsec_retarded()`); the dipole bound-free machinery of Sections 3.4–3.7 is retained as the legacy and validation path. The same- n fine-structure M1 rate uses the analytic magnetic-dipole formula Eq. (21) (`fine_structure_m1()`), because the retarded integral is numerically unstable at the tiny fine-structure photon momentum.

All bound-bound rate functions and the RR cross section guard against non-existent states with $l \geq n$ (e.g. a $1p$ state) by returning zero, using the `valid_kappa()` condition. The rate functions accept a state (n, κ) only if $l(\kappa) < n$, so the combined rate A_{total} vanishes for any pair involving such a state.

Function	Purpose
<code>retarded_total()</code>	Fully retarded plane-wave rate (s^{-1}); basis functions P, Q ; M1
<code>fine_structure_m1()</code>	Analytic same- n M1 rate (s^{-1})
<code>dirac_transrate()</code>	E1 rate (s^{-1})
<code>dirac_e2_rate()</code>	E2 rate (s^{-1})
<code>dirac_m1_rate()</code>	M1 rate (s^{-1})
<code>dirac_total_transrate()</code>	Combined E1+E2+M1 rate (s^{-1})
<code>dirac_lifetime()</code>	Lifetime $\tau_{n\kappa}$ (s) including all multipoles
<code>dirac_branch()</code>	Branching ratio $\tau \times A_{\text{E1}}$ (E1 only)
<code>valid_kappa()</code>	Checks $l < n$ for a given κ
<code>dirac_continuum_PQ()</code>	Energy-normalised continuum Dirac radial integrals P, Q : adaptive DOPRI5 integration with r^γ near-origin representation and envelope asymptotic matching (internal, <code>static</code> to DiracRR.cxx)
<code>dirac_rr_xsec()</code>	Per-vacancy RR cross section via Milne relation (internal, <code>static</code> to DiracRR.cxx); fully-retarded plane-wave path by default, <code>retarded</code> otherwise
<code>dirac_rr_xsec_retarded()</code>	Fully-retarded (plane-wave, all-multipole) per-vacancy RR cross section (cm^2); sums <code>retarded_channel</code> over the $ j_c - j_b = 0, 1, 2, \dots$ shells (internal, <code>static</code> to DiracRR.cxx)
<code>retarded_channel_tot()</code>	Per-channel fully-retarded matrix-element squared $\sum_{m_c, m_b, \epsilon} M ^2$, combining the retarded radial integrals with the <code>ang_int()</code> coefficients (internal, <code>static</code> to DiracRR.cxx)
<code>dirac_rr_bf_retarded(), ..._taylor(), ..._numeric()</code>	Retarded bound-free radial integrals R_+ via closed-form branch sum, small- k_p Taylor series, or quadrature fallback (internal, <code>static</code> to DiracRR.cxx)
<code>sigma_IchiharaEichlerRR()</code>	Tabulated Ichihara & Eichler cross section by Z and n (cm^2)

References

- [1] V.G. Pal’chikov, Phys. Scr. **57**, 581 (1998).
- [2] V.A. Knyazeva, K.N. Lyashchenko, M. Zhang, D. Yu, and O.Yu. Andreev, Phys. Rev. A **106**, 12809 (2022).
- [3] J. Sapirstein, K. Pachucki, and K.T. Cheng, Phys. Rev. A **69**, 022113 (2004).
- [4] O. Jitrik and C.F. Bunge, J. Phys. Chem. Ref. Data **33**, 1059 (2004).
- [5] B. Müller, J. Rafelski, and W. Greiner, Il Nuovo Cimento A **18**, 551 (1973).
- [6] A. Ichihara and J. Eichler, At. Data Nucl. Data Tables **74**, 1 (2000).